

DATA MAPPING DOCUMENT

Purpose

Defines how source data flows into analytical datasets.

Source to Target Mapping

Source	Target	Transformation
students.csv	Student Dimension	Direct load
question_bank.csv	Question Dimension	Direct load
tests.csv	Test Dimension	Direct load
responses.csv	Fact Response	Join with question and test
responses + question_bank	Question Analytics	Calculate accuracy
responses + scores	Student Progress	Aggregate scores
student_progress + students	Risk Report	Apply business rules

Data Transformation Rules

Rule ID	Transformation
TR001	Calculate percentage from correct answers
TR002	Assign grades based on percentage
TR003	Classify risk based on performance
TR004	Calculate chapter accuracy
TR005	Calculate improvement trend

Data Flow

Student Data



Assessment Data



Question Bank



Responses



Python ETL Pipeline



Generated CSV Files



Microsoft SQL Server



SQL Data Validation



SQL Business Analysis



Power BI Dashboard